

Facial Recognition Bias Audit Teacher Notes

TOK Cognitive Science Activity Bank • Knowledge and Technology • 35-45 min

Category	Knowledge and Technology	Time	35-45 min
Difficulty	Medium	ID	facial-recognition-bias-audit

Big idea: Automated recognition is a knowledge claim whose reliability can vary across contexts and groups.

Overview

Students audit a fictional facial recognition system and decide what evidence would justify trusting or limiting it.

Student Prompt

Inspect Facial Recognition Bias Audit Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Facial Recognition Bias Audit Stimulus A	Student-facing stimulus 1: Audit row: hallway camera, 94% overall accuracy, but higher false positives in low light. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Facial Recognition Bias Audit Stimulus B	Student-facing stimulus 2: Stakeholder card: student wrongly flagged, administrator managing safety, developer defending the model. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Use this for comparison, a second round, or a partner challenge.
Facial Recognition Bias Audit Reveal / Twist	Student-facing stimulus 3: Stakeholder card: student wrongly flagged, administrator managing safety, developer defending the model. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Teacher reveal: connect the changed response to the big idea — Automated recognition is a knowledge claim whose reliability can vary across contexts and groups.

Actual Lesson Questions and Key

#	Question for students	Teacher key / cue
1	After inspecting Facial Recognition Bias Audit Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Facial Recognition Bias Audit Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.

#	Question for students	Teacher key / cue
6	When is a technological classification reliable enough to act on?	Teacher cue: students should move from the classroom example to a general TOK issue.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how automated recognition is a knowledge claim whose reliability can vary across contexts and groups.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Setup Checklist

- Use fictional data and avoid using real student faces or sensitive biometric tools.
- Present the system as a claim-making tool: ‘this face matches this identity.’

Ready-to-Use Examples

- Audit row: hallway camera, 94% overall accuracy, but higher false positives in low light.
- Stakeholder card: student wrongly flagged, administrator managing safety, developer defending the model.

Suggested Timing

Stage	Teacher move	Watch for
1	Introduce a fictional school security tool and its claimed accuracy.	Assumptions, confidence shifts, evidence claims, and language choices.
2	Groups inspect audit data broken down by context and subgroup.	Assumptions, confidence shifts, evidence claims, and language choices.
3	Students identify where the headline accuracy hides reliability differences.	Assumptions, confidence shifts, evidence claims, and language choices.
4	Role groups decide whether to deploy, revise, restrict, or reject the tool.	Assumptions, confidence shifts, evidence claims, and language choices.
5	Debrief how technical evidence and ethical responsibility interact.	Assumptions, confidence shifts, evidence claims, and language choices.
Debrief	Move from the activity result to a TOK knowledge question.	Students distinguishing method, evidence, interpretation, and overclaiming.

Teacher Language

- Overall accuracy is not the whole knowledge claim.
- Who bears the cost when a technological knower is wrong?

Common Misconceptions

- Students may treat a tool's output as neutral because it feels automatic.
- Students may focus only on user error and miss design incentives or hidden rules.

Assessment Look-Fors

- Identifies the rule, design choice, or incentive shaping knowledge.
- Distinguishes access to information from justified knowledge.

Differentiation and Adaptations

- Support: use two contexts and one error type.
- Challenge: ask groups to set a minimum evidence threshold for deployment.

Extension Tasks

- Students draft a transparency notice for any automated classification system.
- Compare algorithmic recognition with human eyewitness recognition.

Related Activities

- AI Hallucination Courtroom
- Deepfake Detection Challenge
- What Counts as Evidence?